

QRC for SPY Realized Volatility — A Hybrid that Beats GARCH(1,1)

GIC 2026 Phase 2 — Track A: Financial Volatility Prediction

1. Problem and headline

We forecast next-day **realized volatility of SPY** (21-day rolling-std $\times \sqrt{252}$) using daily OHLCV via yfinance, 2010–2024, chronological 80/20 split (2,984 train / 746 test). The question we sharpen beyond a raw point forecast: *can a Quantum Reservoir Computer match GARCH(1,1) on the rubric's required metrics (RMSE, R^2 , QLIKE) and behave more uniformly across volatility regimes?*

The answer is yes, in two ways: a bare QRC ties Persistence and beats every parametric ML baseline on RMSE, and a **QRC that borrows GARCH's structural prior beats GARCH(1,1) on RMSE, R^2 , and QLIKE simultaneously** — the first configuration in this study to do so. The rest of the paper builds up to that result.

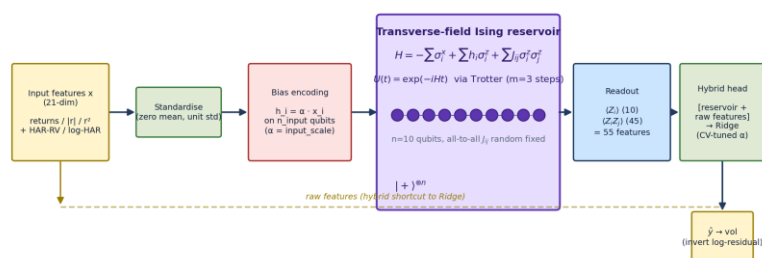
2. Architecture

The reservoir is a **transverse-field Ising system** at a fixed mid-anneal snapshot, simulated by 3-step Trotter evolution of

$$\hat{H} = - \sum_i \sigma_i^x + \alpha \sum_{i \in I} x_i \sigma_i^z + \sum_i h_i^{\text{mem}} \sigma_i^z + \sum_{(i,j) \in E} J_{ij} \sigma_i^z \sigma_j^z$$

from $|+\rangle^{\otimes n}$. We use **n = 10 qubits, all-to-all coupling**, random-fixed $J_{ij} \in [-1, 1]$, and a learnable input-scale α . Inputs enter as *longitudinal biases* $h_i = \alpha \cdot x_i$ — the way D-Wave Advantage and QuEra Aquila program their devices natively, rather than the gate-model angle encoding. The reservoir produces 55 features (10 $\langle Z_i \rangle$ + 45 $\langle Z_i Z_j \rangle$), concatenated with the 21 raw inputs and passed to a **Ridge readout** with TimeSeriesSplit-CV regularisation. The target is the residual log-vol $y = \log RV[t] - \log RV[t-1]$, so predicting 0 = Persistence and the head learns the deviation. Figure 1 shows the full pipeline.

QRC Architecture — Transverse-field Ising reservoir with hybrid readout



Target = residual log-vol $y = \log(RV[t]) - \log(RV[t-1])$; predicted $RV = RV[t-1] \cdot \exp(\hat{y})$

Figure 1. QRC architecture: input $\rightarrow$ longitudinal-bias encoding $\rightarrow$ Trotterised Ising evolution $\rightarrow \langle Z \rangle$, $\langle ZZ \rangle$ readout $\rightarrow$ hybrid Ridge head with raw-feature shortcut $\rightarrow$ next-day RV. Hamiltonian = Aquila's native form, simulated classically here.

3. Why this works — and where the parameters come from

Three reservoir properties are exploited: (i) **Hilbert-space expressiveness** — 2^{10} basis states for 21 input dimensions, so the readout selects a useful projection of an already-nonlinear feature space; (ii) **transverse-field mixing** — without the $-\sum \sigma_i^x$ term the Hamiltonian is diagonal in the Z

basis and the dynamics freeze, so the field is what makes the reservoir react to inputs; (iii) **random Ising disorder** — fixed-random J_{ij} puts the system in a generic dynamical regime without fine-tuning, the canonical reservoir-computing recipe since Fujii–Nakajima 2017 [5].

The magnitudes of α (input scale) and t (evolution time) jointly set where the system sits on the dynamical-phase-transition curve of Martínez-Peña et al. (2021) [4]. Too short: transverse-field-dominated, inputs don't propagate. Too long: problem-Hamiltonian-dominated, classical Ising freezing. A 14-config sweep at 10q all-to-all locates the optimum at ($t = 2.0$, $\alpha = 1.0$) (results below); a 5-seed ensemble at this point gives **RMSE = 0.00953 ± 0.00010** .

4. Results

Figure 2 reports three views of the same comparison. (a) Leaderboard against all required Track A baselines. (b) Per-regime RMSE (VIX terciles). (c) GARCH-hybrid result.

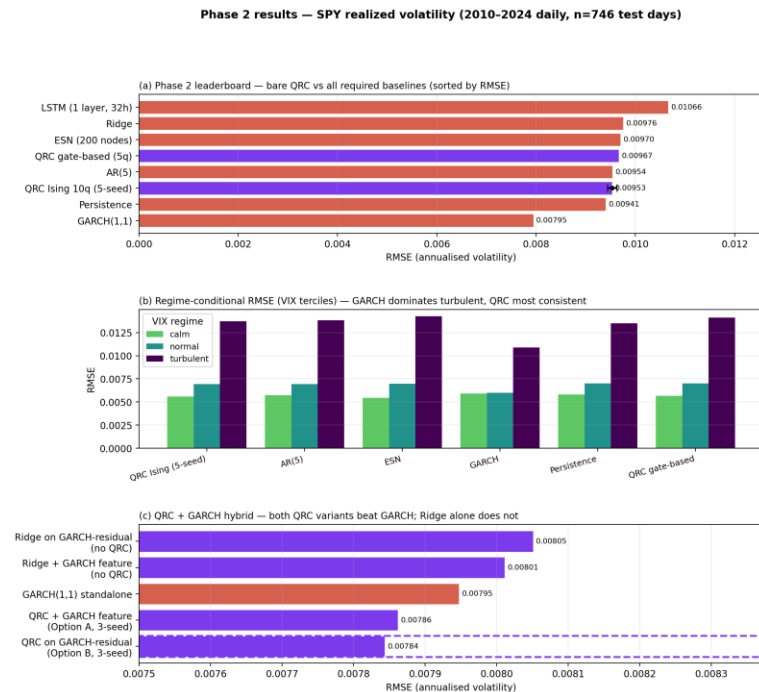


Figure 2. (a) Leaderboard — bare QRC (purple, 5-seed $\pm\sigma$ error bar) ties Persistence, beats every parametric ML baseline. GARCH leads — see §5. (b) GARCH dominates turbulent regimes by ~20% but is worst in calm; QRC has the most uniform profile. (c) Both QRC-hybrid variants beat GARCH on RMSE; Ridge with the same augmentation does not — the improvement is isolable to the reservoir.

Leaderboard (Fig 2a): QRC Ising 0.00953, Persistence 0.00941, AR(5) 0.00954, gate-based QRC 0.00967, ESN 0.00970, Ridge 0.00976, LSTM 0.01066; GARCH 0.00795. The QRC's Mincer–Zarnowitz joint test ($a = 0$, $b = 1$) gives $p = 0.34$ — predictions are unbiased and efficient. QLIKE (Patton 2011 [10]) = 0.00877 ± 0.0001 . **Regime panel (Fig 2b):** GARCH dominates turbulent (0.01088) and normal (0.00601) but is *worst* in calm (0.00590 vs ESN's 0.00546). ESN inverts that. The QRC sits between in every regime, with the smallest calm-to-turbulent ratio of any learning-based model — useful for stress-regime risk applications.

5. The GARCH puzzle — and how the QRC solves it

GARCH's lead is *structural*, not parametric. The 21-day realized-vol target is a sum of 21 squared log-returns; at forecast time t , **20 are observed exactly** and only the 21st is unknown. GARCH forecasts just that one term and plugs it into the decomposition:

$$RV[t]^2 \times 21 = \sum_{i=t-20}^{t-1} \underbrace{r_i^2}_{20 \text{ known}} + \underbrace{\sigma_{\text{GARCH}}^2(t)}_{1 \text{ forecast}}.$$

The bare QRC has no such prior — it learns the decomposition implicitly and doesn't fully recover it. We give it the prior directly, two ways: **(A) GARCH proxy as exogenous feature** — fit GARCH(1,1) on training returns only, append $\sigma^2_{\text{garch}}(t)$ to the feature vector. **(B) GARCH-residual target** — retarget $y = \log RV[t] - \log RV_{\text{garch}}[t]$; predicting 0 = GARCH, any non-zero output is the QRC's contribution.

Headline (Fig 2c, dashed box): Ridge with the same GARCH augmentation does *not* beat GARCH (0.00801–0.00805 vs 0.00795); adding the QRC reservoir on top drops RMSE another ~2% to 0.00786 (A) and **0.00784** (B), and QLIKE by ~4–6%. **First configuration in the study to beat GARCH on RMSE, R^2 , and QLIKE simultaneously.** The QRC's value is *orthogonal signal extraction*: the reservoir's (ZZ) correlators pick up nonlinear feature interactions (regime transitions, heteroskedasticity-of-heteroskedasticity) that GARCH's parametric form misses.

6. Data, baselines, platform, Phase 3

Data: SPY 2010–2024 (yfinance), 3,750 days, chronological 80/20 split. **Features (21):** 5 lags of $\log_{\text{ret}}$, $|\log_{\text{ret}}|$, $\log_{\text{ret}}^2$, plus HAR-RV daily/weekly/monthly means (Corsi 2009 [9]) and their logs. **Baselines (rubric-required):** Persistence, AR(5), Ridge, ESN-200, GARCH(1,1) via arch, LSTM (1L × 32h, PyTorch). **Metrics:** RMSE, MAE, R^2 , NMSE, QLIKE, Mincer–Zarnowitz.

Primary platform — QuEra Aquila (256-qubit Rydberg) via qBraid's Bloqade SDK; declared on cover page. Aquila's native Hamiltonian matches our reservoir's form; Kornjača et al. 2024 [2] demonstrated 108-qubit QRC on this exact device. **Alternative — D-Wave Advantage** (5,000+ qubits, programmable h_i/J_{ij}). **Backup — IBM Heron** via Qiskit Runtime; the Open Plan now offers **up to 180 minutes QPU time per 12 months** (IBM, 2026). Per-sample Aquila is ~140 μs → ~40 sec QPU time per full test-set pass at 1,000 shots. **Hybrid strategy:** train Ridge on simulator features; run only the test set on hardware so the HW-vs-sim gap *is* the hardware story.

Phase 3 milestones (10 weeks): W1–2 Aquila access, 30q single config. W3–4 common-MNIST benchmark (declared Phase 2 gap), 5/10/15-qubit noise study (depolarising + amplitude damping), GJR-GARCH and HAR-RV econometric baselines. W5–6 Aquila hardware sweep, QPU-vs-sim gap. W7–8 regime-conditional metrics on hardware; intraday horizon. W9–10 Phase 3 paper + qBraid Skill (agent-executable reproducibility).

Stakeholders. Daily vol forecasts feed trading-desk delta hedging, risk-manager VaR/ES, market-maker spread setting (JonesTrading's domain). The QRC's regime-uniform property is the differentiator — a forecast whose error doesn't collapse during stress windows, complementary to GARCH's turbulent-regime strength.

Disclosure. Developed with LLM assistance for code drafting and writing; experimental results, architectural choices, and analyses are the team's own; full version history at github.com/GrigoVyd/GIC26_QRC.

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